

Solving by Substitution Notes

Example 1: Solve the system of linear equations by substitution

$$\begin{cases} 2x - 3y = 13 \\ x - 4y = 9 \end{cases}$$

$x = 9 + 4y$
 $x = 9 + 4(-1)$
 $x = 5$

$2(9 + 4y) - 3y = 13$
 $18 + 8y - 3y = 13$
 $18 + 5y = 13$
 $5y = -5$
 $y = -1$

(5, -1)

3 × 3 Substitution

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3 × 3 systems of linear equations have three variables and three equations.

The solution to a three variable system of equations may be represented by an ordered triple.

Process

Steps to Solve 3 × 3 Systems of Linear Equations by Substitution

1. Isolate a variable in one equation.
2. Substitute the expression representing that variable into both of the other equations to create a new 2 × 2 system.
3. Solve the new 2 × 2 system for both of its variables.
4. Substitute the values found in step 3 into any of the original equations to solve for the remaining variable.
5. Write the solution as an ordered triple, if appropriate.

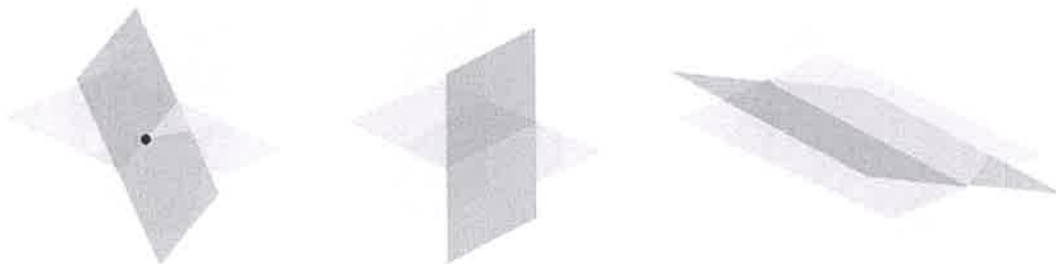
Special Solutions

Key Concepts

3 × 3 systems of equations can have solutions at a point, such as (x, y, z) . This occurs when the three planes (represented by the three equations) intersect at one location.

If the system has no solution, one of the variables will eliminate, resulting in a false statement, such as $0 = 1$. This occurs when there is no location where all three planes intersect.

If the system has infinite solutions, one of the variables will eliminate, resulting in a true statement, such as $0 = 0$. This occurs when the planes intersect in a line or when the planes are stacked exactly on top of each other.



Example 2: Solve the system of linear equations by substitution

$$\begin{cases} a+b+c=6 \\ 2a-b+3c=-59 \\ a-b-c=-12 \end{cases}$$

$$\begin{aligned} a+b+c &= 6 \\ -a-b & \quad -a-b \\ c &= 6-a-b \\ c &= 6-(-3)-(-20) \end{aligned}$$

$$\begin{aligned} 2a-b+3c &= -59 \\ 2a-b+3(6-a-b) &= -59 \\ \underline{2a} \quad \underline{-b} \quad +18 \quad \underline{-3a} \quad \underline{-3b} &= -59 \\ -a-4b+18 &= -59 \\ \quad \quad \quad \underline{-18} \quad \quad \underline{-18} \\ -a-4b &= -77 \\ -(-3)-4b &= -77 \\ \quad \quad \quad \underline{3} \quad \underline{-4b} &= -77 \\ \quad \quad \quad \underline{-3} \quad \quad \underline{-3} \\ \quad \quad \quad -4b &= -80 \\ \quad \quad \quad \underline{-4} \quad \quad \underline{-4} \end{aligned}$$

$$b = 20$$

$$\begin{aligned} a-b-c &= -12 \quad c=3-14=-11 \\ a-b-(6-a-b) &= -12 \\ \underline{a} \quad \underline{-b} \quad -6 \quad \underline{+a} \quad \underline{+b} &= -12 \\ 2a-6 &= -12 \\ \quad \quad \underline{+6} \quad \quad \underline{+6} \\ \underline{2a} &= -6 \\ \quad \quad \underline{2} \quad \quad \underline{2} \\ a &= -3 \end{aligned}$$

a, b, c

$$\boxed{(-3, 20, -11)}$$

Example 3: Solve the system of linear equations by substitution

$$\begin{cases} x-2y-z=3 \\ y+3z=6 \\ 2x-3y+z=9 \end{cases}$$

$$\begin{aligned} x-2y-z &= 3 \\ +2y+z &+2y+z \\ x &= 3+2y+z \end{aligned}$$

$$\begin{aligned} 2(3+2y+z)-3y+z &= 9 \\ \underline{6} \quad \underline{+4y} \quad \underline{+2z} \quad \underline{-3y} \quad \underline{+z} &= 9 \\ y+3z+6 &= 9 \\ \quad \quad \quad \underline{-6} \quad \underline{-6} \\ y+3z &= 3 \end{aligned}$$

$$\begin{aligned} -1(y+3z) &= -6 \\ y+3z &= 3 \\ \underline{-y-3z} &= -6 \\ 0x-3 & \end{aligned}$$

No Solution

Example 4: Solve the system of linear equations by substitution

$$\begin{cases} 2a + b - c = 5 \\ 4a + 2b - 2c = 10 \\ a + b + 3c = 15 \end{cases}$$

$$\begin{aligned} 4a + 2b - 2c &= 10 \\ -4a &\quad +2c \quad -4a + 2c \\ \hline 2b &= 10 - 4a + 2c \\ \frac{2b}{2} &= \frac{10}{2} - \frac{4a}{2} + \frac{2c}{2} \\ b &= 5 - 2a + c \end{aligned}$$

$$2a + (5 - 2a + c) - c = 5$$

$$\underline{2a} + 5 - \underline{2a} + \underline{c} - \underline{c} = 5$$

$$5 = 5$$

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Try This! Summarize the steps for solving a system of linear equations by substitution.

Compare and contrast solving 3×3 linear systems by substitution and solving 2×2 systems by substitution.

Sample 2×2 system: $\begin{cases} 2x - 3y = 13 \\ x - 4y = 9 \end{cases}$

Sample 3×3 system: $\begin{cases} a + b + c = 6 \\ 2a - b + 3c = -59 \\ a - b - c = -12 \end{cases}$